

## E.1 / PAPER 1 / QUESTION 20

3 November 2022 / TZ0 / HL Paper 1 / original Q40 / 8822-6501

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20. The nucleus of the isotope hydrogen-2 has a radius  $R$  and a density  $\rho$ .

What are the approximate radius and density of a nucleus of oxygen-16?

|    | Oxygen-16<br>radius | Oxygen-16<br>density |
|----|---------------------|----------------------|
| A. | $2R$                | $\rho$               |
| B. | $2R$                | $2\rho$              |
| C. | $8R$                | $\rho$               |
| D. | $8R$                | $2\rho$              |

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References:

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